> I'm analyzing a dataset of YouTube videos ranked by comment count. I've compared the 10th, 100th, and 1000th most-commented videos. What kinds of patterns or trends should I look for in the comment counts as I move further down the rankings? Can you suggest any questions I should ask myself to better understand the differences between these rankings?

OpenAI - GPT 5.6 Luna ($$)

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As you move from the 10th to the 100th and 1,000th most-commented videos, look for whether comment counts follow a **long-tail or heavy-tailed distribution**. In such distributions, a small number of videos may receive a disproportionately large share of attention, while counts decline rapidly and then flatten among lower-ranked videos. Related video research finds that popularity and attention are often highly uneven: top videos can capture a large fraction of total views, and engagement distributions may be strongly skewed arxiv.org. Algorithmic recommendation can intensify this "rich-get-richer" pattern by repeatedly exposing already successful content to more users knightcolumbia.org.

Patterns to examine

1. **Absolute decline**
   - How many comments separate ranks 10 and 100?
   - How many separate ranks 100 and 1,000?
   - Is the decline approximately linear, exponential, or closer to a power law?
2. **Rank ratios**

   Compare ratios such as:

   $$\frac{C_{10}}{C_{100}}, \qquad \frac{C_{100}}{C_{1000}}, \qquad \frac{C_{10}}{C_{1000}}$$

   where C_r is the comment count at rank r. Large ratios indicate strong concentration near the top.
3. **Log-log rank plot**

   Plot rank on the x-axis and comment count on the y-axis using logarithmic scales. A roughly straight declining line may suggest a power-law-like rank distribution; curvature may indicate that different parts of the ranking behave differently.
4. **Percentage of total comments**

   Calculate how much of all comments are generated by:
   - the top 10 videos,
   - the top 100,
   - the top 1,000,
   - the remaining videos.

   This is often more informative than comparing individual ranks. For example, the top 1% may account for a very large share of engagement even if the exact distribution is not a pure power law.
5. **Changes in slope**

   The gap between ranks 10 and 100 may be much larger than the gap between 100 and 1,000, or vice versa. A "bend" in the curve could indicate separate groups, such as viral videos, established channels, and long-

tail content.

6. **Comment counts relative to views**

   Raw comments primarily measure accumulated attention, not necessarily discussion intensity. Calculate:

   $$\text{comment rate} = \frac{\text{comments}}{\text{views}}$$

   A video with many comments may simply have many more viewers. Research on video engagement similarly distinguishes popularity measures such as views from deeper engagement measures such as watch time and watch percentage arxiv.org.

7. **Comments relative to channel size**

   Compare comment counts with subscribers, channel views, upload frequency, or the creator's typical comment count. This can reveal whether lower-ranked videos are unexpectedly strong performers for smaller creators.

8. **Age and exposure**

   Older videos have had more time to accumulate comments. Consider comparing videos by upload year or measuring comments per day since upload.

9. **Content and audience differences**

   Examine whether the top-ranked videos differ from lower-ranked videos in:

   - category or topic,
   - video length,
   - channel size,
   - language or geography,
   - live-stream versus prerecorded format,
   - controversial or news-related content,
   - calls to comment or questions in the video.

10. **Outliers and anomalies**

    Identify videos whose comment counts are unusually high relative to their views. These may reflect strong community engagement, controversial content, coordinated activity, spam, or unusual moderation histories.

Questions to ask

- Is the distribution dominated by a few extreme videos?
- Does the top 10 represent a qualitatively different group from the top 100 or top 1,000?
- Are the differences mostly caused by views, or do comment rates also differ?
- Does the ranking remain similar when using comments per 1,000 views?
- Are comments concentrated among a few channels?
- Do some categories consistently produce higher comment rates?
- Are older videos overrepresented near the top?
- Does video length correlate with comment count after controlling for views?
- Are highly commented videos also highly watched, or are comments and watch time ranking videos differently?
- How sensitive are the results to deleted comments, disabled comments, spam filtering, or videos with

unusually large audiences?

- Does the distribution change across time periods or YouTube categories?
- Would the conclusions change if you analyzed medians and percentiles rather than only ranks 10, 100, and 1,000?

A useful first analysis would be a table containing, for each cutoff, the comment-count threshold, cumulative comments, percentage of total comments, median views, median comment rate, median channel size, and median video age. This would show whether the ranking reflects broad popularity, unusually active audiences, or both. Note that the provided research supports analyzing skew, concentration, and the distinction between popularity and engagement, but it does not provide a universal expected comment count for ranks 10, 100, or 1,000.

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